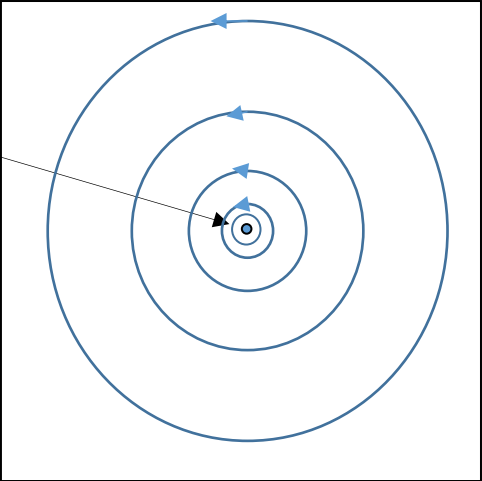


4	(a)(i)	Wire X with current flowing out of the plane.  Magnitude of magnetic field strength: Increasing with separation of the concentric circles from the centre at wire X Direction of magnetic field: Anticlockwise direction based on right hand grip rule	B1 B1
	(a) (ii)	Since the currents of the parallel current-carrying wires are the same in direction, the magnetic forces are attracting each other. Hence, the direction of the magnetic force is from Y towards X.	B1

	(b)(i)	B at wire Y due to wire X, $B = \frac{\mu_0 I_X}{2\pi x}$ $B = \frac{4\pi \times 10^{-7} \times 5.0}{2\pi \times 2.5 \times 10^{-2}}$ $= 4.0 \times 10^{-5} \text{ T}$ $F = B I_Y L \sin \theta$ $F = 4.0 \times 10^{-5} \times 7.0 \times L \sin 90^\circ$ $\frac{F}{L} = 2.8 \times 10^{-4} \text{ N}$	C1 C1 A1
	(b)(ii)	The magnetic force acting on wire X due to wire Y and the magnetic force acting on wire Y due to wire X are a pair of action and reaction forces. According to Newton's 3rd law, the forces on the two wires are equal in magnitude. <u>OR</u> The magnitude of the magnetic force depends on the product of the currents in the two wires. Hence the forces on the two wires are equal in magnitude.	M1 A1 M1 A1
	(c)	$F_Y = B_Y I_X L \sin \theta$ $F_Y = \frac{\mu_0 I_X \sin \omega t}{2\pi x} \times I_Y \sin \omega t \times L \sin \theta$ $F_Y = \frac{\mu_0 L I_X I_Y}{2\pi x} \times \sin^2 \omega t$ Since F_Y is proportional to $\sin^2 \omega t$, the variation of the magnitude of the magnetic force is square of sinusoidal (or sinusoidal) OR the variation of the magnitude of the magnetic force is twice the frequency of the current. Since the currents are in phase, i.e. the currents in the two wires are always in the same direction, the magnetic forces acting on the wires are always attractive, (except when the currents in the wires are both zero). The direction of the magnetic force is always towards wire X (or leftwards or always in the same direction). (One mark to describe the magnitude and One mark to describe the direction)	B1 B1

5	(a)	The value of voltage that is equivalent to that of a steady d.c. voltage which	P1
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